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Du 24 au 28 mai

Safe Reinforcement Learning Control via Offset-Free MPC Pre-Training and a Data-Driven Lyapunov Safety Layer

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Presentation Outline

- ➊ **Introduction:** MPC, reinforcement learning, and safety motivation
- ➋ **Methodological background:** offset-free MPC pretraining and Lyapunov-based safety
- ➌ **Proposed framework:** RL action proposal, Lyapunov gate, and fallback MPC
- ➍ **Case study:** styrene polymerization CSTR
- ➎ **Results and discussion**
- ➏ **Conclusions and future work**

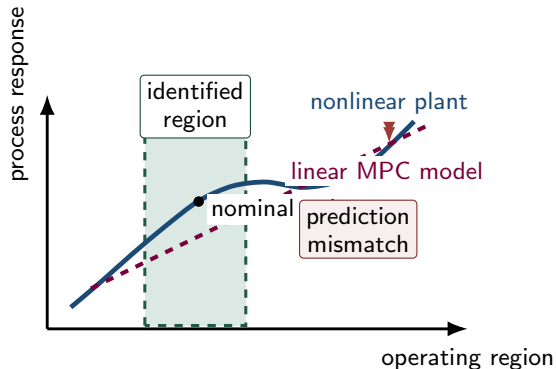
Linear MPC Is Practical, But Its Model Is Local

Why MPC is a strong baseline

- Handles MIMO interactions and constraints.
- Solves a familiar receding-horizon optimization problem.
- Offset-free augmentation helps reject steady disturbances.

Main limitation

- The fixed linear model is local to the identified region.
- Away from nominal operation, mismatch can degrade tracking.



Linear prediction is useful near the nominal region, but mismatch grows away from it.

(Qin and Badgwell, 2003; Mayne et al., 2000)

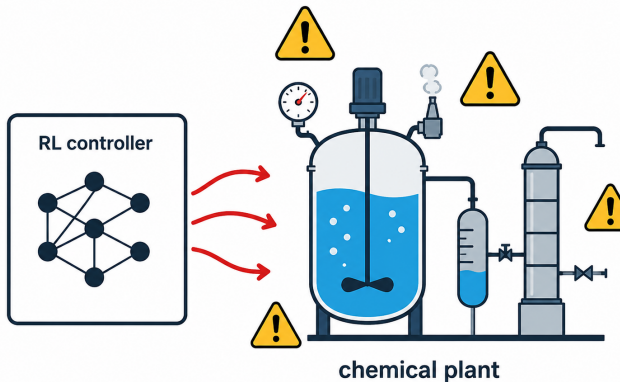
RL Can Adapt, But Online Exploration Is Risky

RL opportunity

- Learns from process data.
- Adapts to model mismatch.
- Can improve beyond MPC.

Plant-safety issue

- Exploration can test unsafe inputs.
- Unsafe inputs can violate constraints.
- Actions must be checked before actuation.



Direct online RL can send risky candidate actions to the plant unless a safety layer intervenes.

(Sutton and Barto, 2018; Garcia and Fernandez, 2015)

Imitation-Learning Pretraining Warm-Starts the RL Actor

Idea

- Use OF-MPC as an expert controller in offline mode.
- Collect expert transition tuples.
- Initialize the actor by imitation before online RL.

Behavior-cloning loss

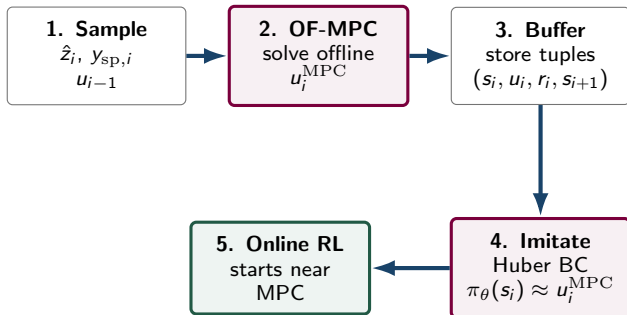
$$s_i = [\hat{z}_i, y_{\text{sp},i}, u_{i-1}]$$

$$\tau_i = (s_i, u_i^{\text{MPC}}, r_i, s_{i+1}), \quad \mathcal{D}_{\text{IL}} = \{\tau_i\}_{i=1}^N$$

$$e_i = \pi_{\theta}(s_i) - u_i^{\text{MPC}}$$

$$\mathcal{L}_{\text{BC}}(\theta) = \frac{1}{N} \sum_{i=1}^N \text{Huber}_{\delta}(e_i)$$

(Hassanpour et al., 2024; Hamed et al., 2026)



Warm start: the actor begins from an MPC-like policy instead of random actions.

Purpose of this step

Pretraining does not replace online learning; it gives the actor a practical starting policy before nonlinear-plant interaction.

Pretraining Gives a Good Start, But Online Safety Is Still Uncertified

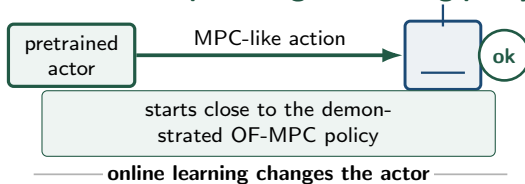
What pretraining gives

- Less random early exploration.
- Starts the actor near a known controller.
- Makes online RL more implementable.

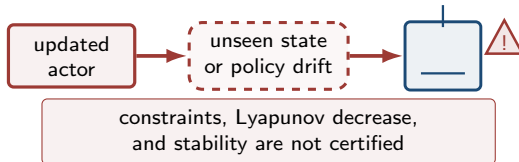
What remains uncertified

- Safety in unseen operating regions.
- Hard input-output constraint satisfaction.
- Lyapunov decrease and closed-loop stability.
- Safe behavior after online actor updates.

Before online updates: good starting policy



After online updates: action may be uncertified



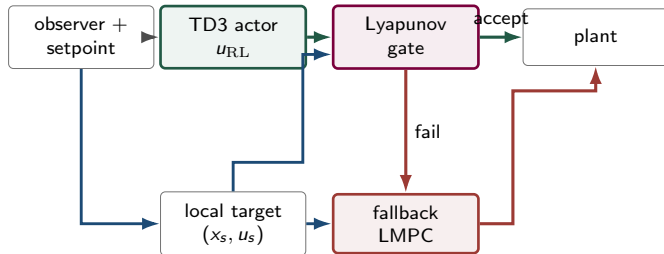
RL Proposes the Move; the Lyapunov Layer Has Final Authority

Control split

- **RL actor:** proposes a performance move.
- **Safety gate:** checks Lyapunov contraction.
- **Fallback LMPC:** replaces rejected actions.

Key design choice

The learned policy is not trusted directly on the plant. The supervisor chooses the final input after certification.



The plant receives either the accepted RL action or the certified fallback action.

Framework: RL is a proposal generator; certification decides what is actuated.

Safety Check: One-Step Lyapunov Contraction

Gate calculation

$$e_k = \hat{x}_k - x_s, \quad V_k = e_k^\top P_x e_k$$

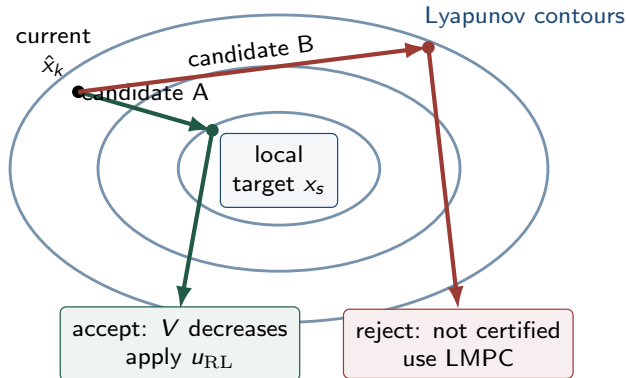
$$V_{k+1}^{\text{cand}} \leq \rho V_k + \epsilon_{\text{lyap}}, \quad 0 < \rho < 1$$

Decision

- **Pass:** apply the candidate RL action.
- **Fail:** replace it with fallback LMPC.

Interpretation

The candidate is accepted only if the predicted next state moves toward the local target in the Lyapunov metric.



Conceptual figure: accepted actions move the state to a lower Lyapunov contour.

Rejected RL Actions Are Replaced by a Lyapunov MPC Move

Two pieces are needed

- **Target selector:** computes a feasible local equilibrium from the current estimate and setpoint.
- **Fallback LMPC:** computes a tracking move that satisfies the contraction condition.

Compact fallback problem

$$\begin{aligned} \min \quad & J_{\text{track}} + J_{\text{move}} \\ \text{s.t.} \quad & u_{\min} \leq u \leq u_{\max}, \\ & V_{1|k} \leq \rho V_k + \epsilon_{\text{lyap}}. \end{aligned}$$

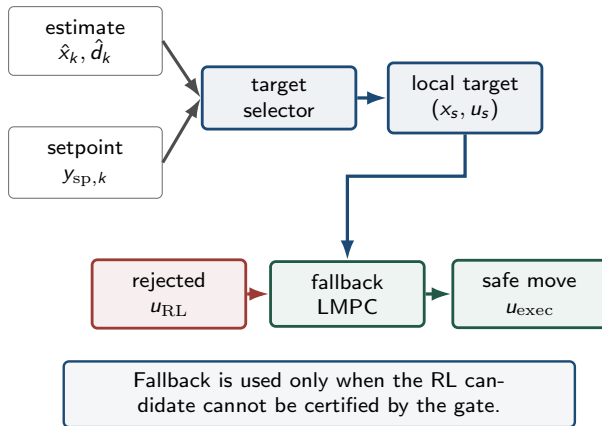


Figure: the target defines the reference for both the gate and fallback controller.

Reward Converts Tracking and Safety Feedback Into Learning

Role of the reward

The reward is the scalar feedback used to update the actor. It should favor actions that improve performance and reduce the need for safety correction.

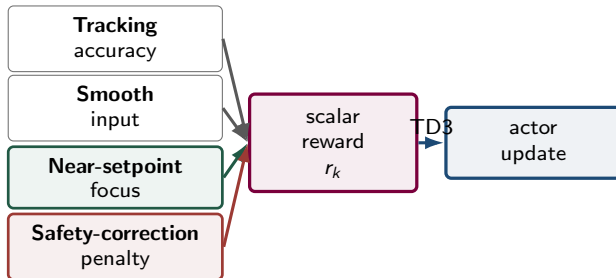
What the reward should encourage

- small output tracking error,
- smooth input movement,
- stronger attention near the setpoint,
- fewer safety corrections by the supervisor.

Interpretation

The actor is not rewarded only for tracking. It is also pushed toward actions that are easier for the safety layer to accept.

Reward construction



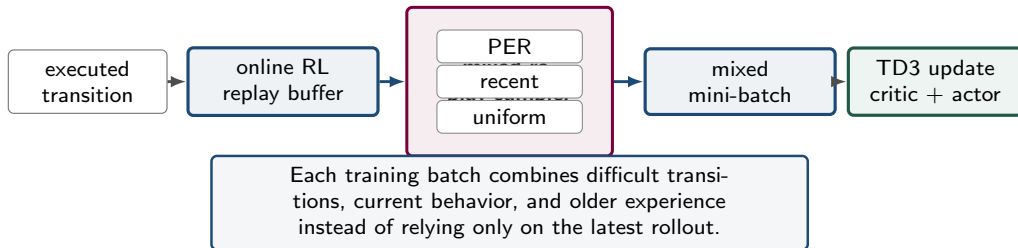
Learning goal: maintain tracking while proposing actions that require fewer safety corrections over training.

Figure: the reward combines performance, smoothness, near-setpoint accuracy, and safety-intervention feedback.

Replay Sampling Balances Rare Failures, Recent Data, and Coverage

Main idea

Online RL should not learn only from the latest trajectory. Mixed replay improves robustness by combining rare high-error events, the current operating regime, and broad historical coverage.



Why three replay views?

PER

revisits high-error and rare transitions

Recent

keeps updates tied to the current regime

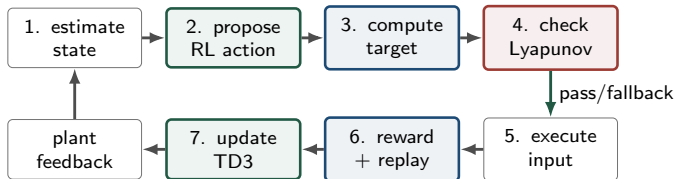
Uniform

preserves broad operating coverage

Online Procedure: Observe, Propose, Certify, Learn

At each sampling time

- Update state and disturbance estimates.
- Generate an RL candidate action.
- Certify using Lyapunov contraction.
- Execute accepted RL or fallback LMPC.
- Store the transition and update TD3.

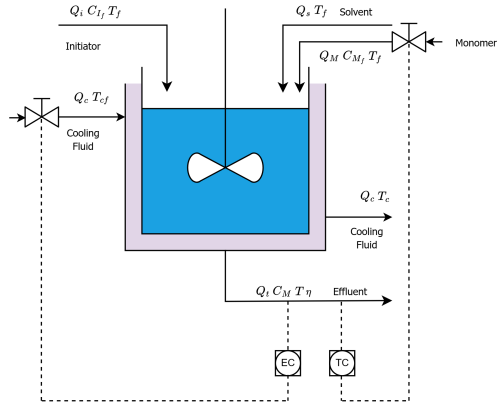


The safety decision is made before actuation; learning happens after the executed transition is observed.

BC window: during the early online phase, the actor softly clones LMPC actions. This keeps the policy close to a certified controller while the replay buffer fills and the actor transitions to full TD3 updates.

Main takeaway: the online controller is an RL proposal embedded inside a certification-and-fallback loop.

Case Study: Styrene Polymerization CSTR



Process schematic: jacketed polymerization CSTR with feed and coolant manipulation.

Why this process is a useful benchmark

- Nonlinear reactor dynamics couple polymer quality and temperature.
- Input constraints and model mismatch make unsafe exploration undesirable.

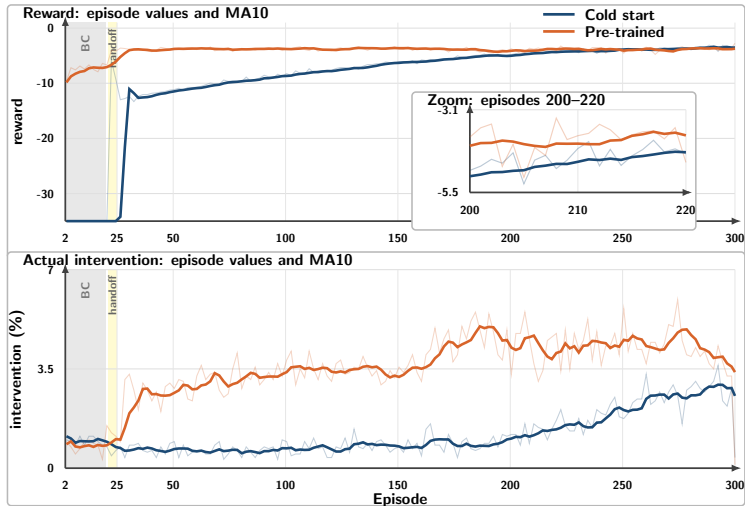
Control setup

Inputs	coolant flow Q_c , monomer flow Q_m
Outputs	intrinsic viscosity η , reactor temperature T
Objective	improve tracking with supervised final action selection

Role in this presentation

Benchmark for fixed MPC, pretrained RL, and Lyapunov-supervised RL.

Reward and Intervention Trends for the Two RL Runs



Summary metrics

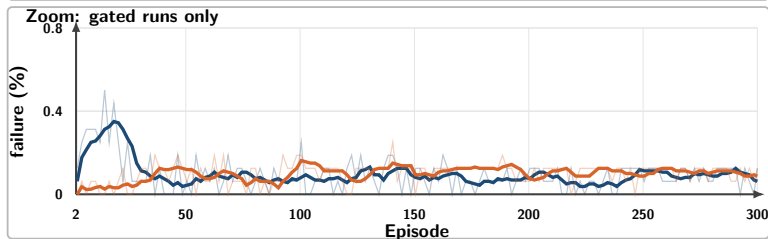
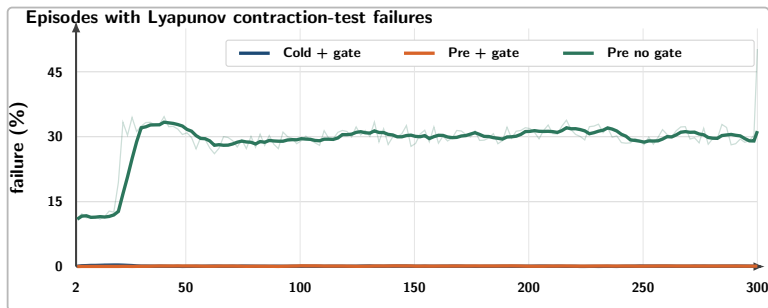
Full run

Metric	Cold	Pre
Reward	-10.38	-4.07
RMSE	0.284	0.185
Interv.	1.19%	3.53%

Last episode

Metric	Cold	Pre
Reward	-3.33	-2.75
RMSE	0.177	0.153
Interv.	0.38%	0.13%

The Lyapunov Gate Enforces a Runtime Certificate



Certificate check

Case	Fail rate
Cold + gate	0.10%
Pre + gate	0.10%
Pre no gate	29.17%

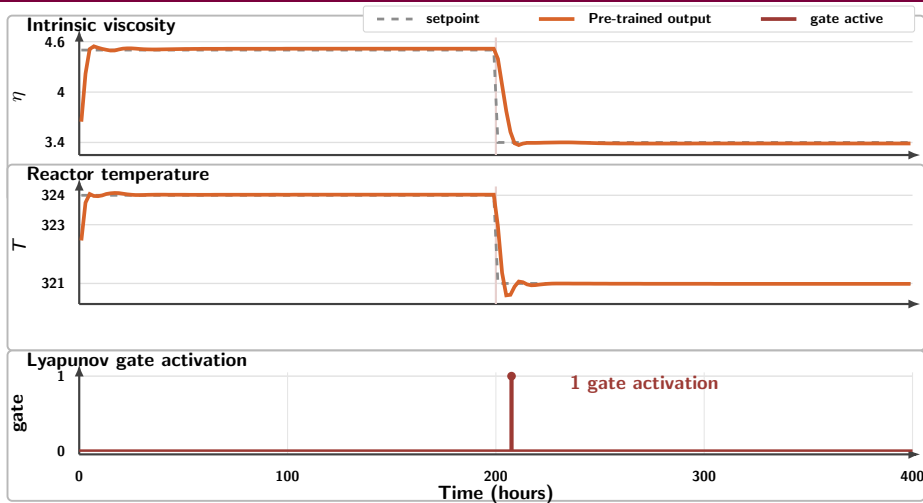
$$\Delta V = V_{k+1} - (\rho V_k + \epsilon)$$

Fail means $\Delta V > 0$.

Interpretation

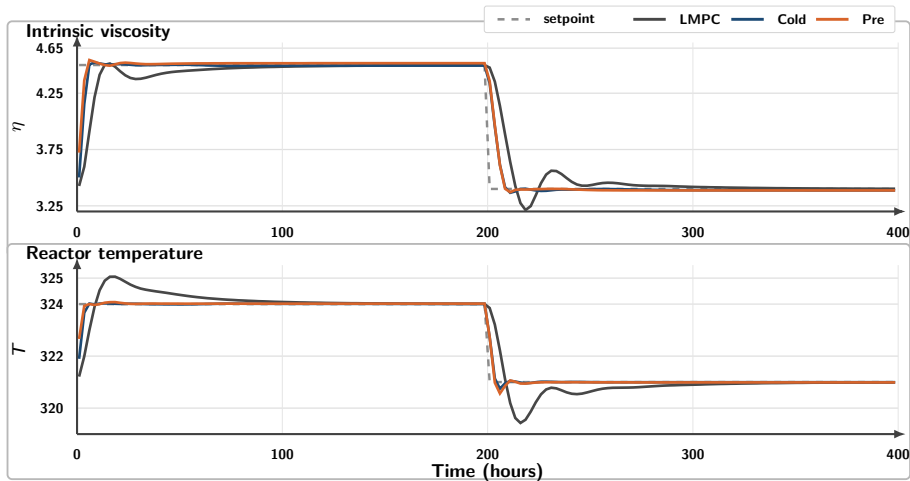
The top panel compares all runs on the same failure-rate scale. The bottom panel is only a zoom of the two gated runs. The gate keeps applied failures near zero; without it, the pre-trained policy often fails the Lyapunov test.

Pre-Trained Evaluation: Tracking With Lyapunov Gate Activation



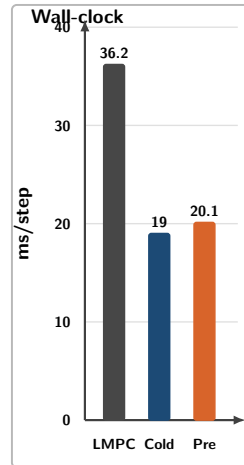
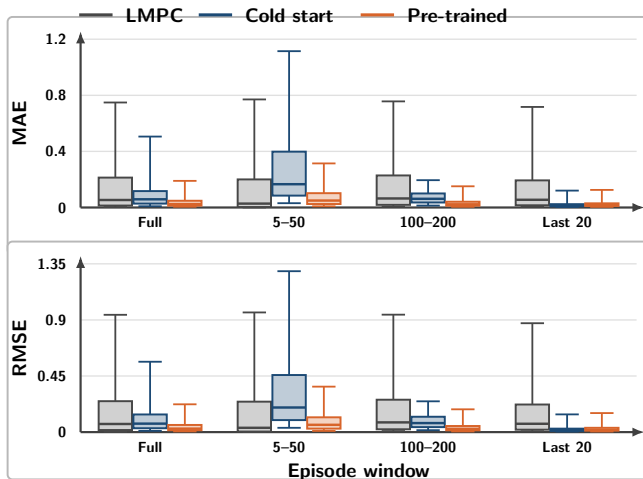
Importance: the gate activates only at the setpoint transition, showing selective certification.

Last-Episode Evaluation Compares All Three Controllers



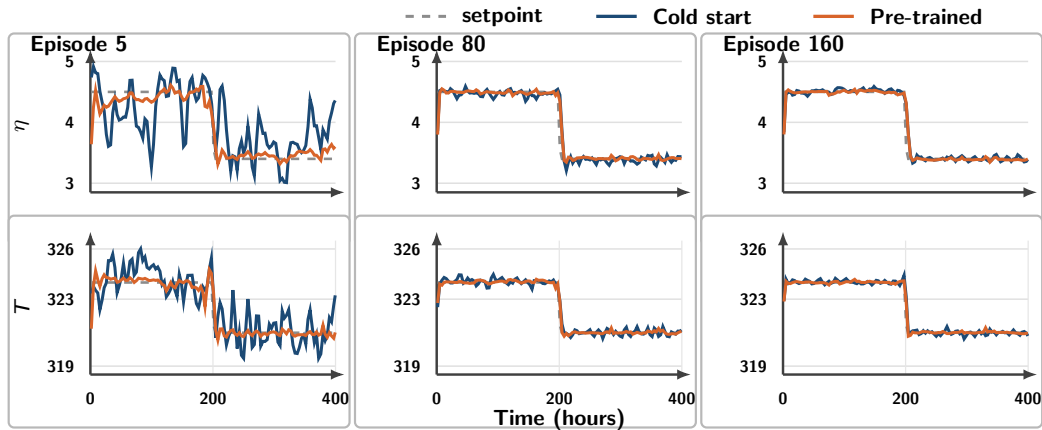
Takeaway: pre-trained gated RL gives the best final-episode tracking through the setpoint transition.

Error Distributions and Wall-Clock Time Across Training Windows



Takeaway: pre-trained RL tightens errors, and both RL runs remain faster than LMPC.

Pre-Training Improves Setpoint Tracking Across Episodes



Interpretation

Across early, middle, and later episodes, the pre-trained agent stays closer to both setpoint trajectories than the cold-start agent, especially after setpoint changes.

Conclusions

- Certified gate for RL actions.
- Practical local guarantee from step-test data and a linear state-space model.
- Pre-training improves tracking; LMPC fallback preserves safe actuation.

Future work

- Pre-train from LMPC demonstrations.
- Scale to a C2-splitter with output constraints and flooding/fault prevention.
- Complete the stability proof.

Thank you